Yin-Chien Pai (Eszter Pai)

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LinkedIn Portfolio GitHub

Continental Automotive

Master Thesis *Berlin* *Feb 2025 - Dec 2025*

Language Conditioned Promptable Traffic Scene Generation

- Developed an automated evaluation loop that boosted scenario diversity by up to 30%, especially in safety-critical cases (e.g., emergency scenarios), leading to broader edge-case coverage in autonomous driving simulation.
- Improved the retrieval module by ~70% through integrating a lightweight LLM-based judge model, significantly increasing the accuracy of map layout selection in the scenario generation pipeline.
- Replaced proprietary LLMs (e.g., Open AI) with open-source alternatives (LLaMA, DeepSeek)

Project

Chat Bot with Zero-Shot Classification, RAG & Multi-Model NLP Pipeline

- Engineered a hybrid classification system combining Toxic-BERT and Zero-Shot classification models, reducing API calls by pre-screening messages before LLM inference.
- Implemented RAG system using ChromaDB to provide context-aware responses, enabling the bot to distinguish between in-game content and real-world toxicity.
- Built a persistent user reputation system with SQLite, tracking user behavior across sessions and dynamically adjusting detection sensitivity based on user history.

Universität Potsdam *Grade 1,7*

Master of Science, Cognitive Systems: Language, Learning and Reasoning *Potsdam* *Sep 2022 - Dec 2025*

Fine-Tuned LLMs for Higher-Order Code Generation in COCOBOTS domain.

- Fine-tuned LLMs, including the LLaMA family and Mistral, for advanced code generation tasks within the COCOBOTS domain.
- Leveraged Unsloth for quantization and LoRA (Low-Rank Adaptation) techniques to optimize parameters and reduce resource requirements.

Designed Dungeon and Dragon Inspired Text Game for Evaluating Chat-Optimized LLMs.

- Evaluated LLM performance based on rule adherence and gameplay intelligence, offering insights into the models' decision-making and contextual understanding.

Other modules taken: Advanced Natural Language Processing, Computer Science, Machine Learning and Data Analysis, Atelier in Experimental and Computational Phonology, Cognitive and Sensorimotor Development.

Hasso-Plattner-Institut

Student Assistant *Potsdam* *Nov 2023 - Jan 2025*

- Created a comprehensive extreme weather dataset to facilitate accurate evaluation and analysis.
- Integrated the extreme weather dataset into a cyclone tracker.

National Cheng-Chi University *Grade 1,3*

Bachelor of Arts, English Language and Literature *Taipei* *Sep 2019 - Aug 2022*

Modules taken: Algorithms, Programming 101, Computational Thinking, Deep Learning Applications, Calculus, Statistics.

Frankfurt School of Finance and Management

Exchange Semester *Frankfurt am Main* *Sep 2021 - Jan 2022*

Skills

Language: Mandarin (Native), English (C1, IELTS 7.5), German (B2).
Skills: Azure Services, Cloud, API, BERT, ChromaDB, Docker, Fine-Tuning, HPC, Latex, Linux, LLMs, LoRA, matplotlib, NLTK, NLP, NumPy, Ollama, OpenAI, Pandas, Prompt Engineering, Python, PyTorch, RAG, Regex, SQLite, scikit-learn, Sloth, SpaCy, TensorFlow, Transformers, Zero-Shot